

SimScale Dynamic Testing Procedure

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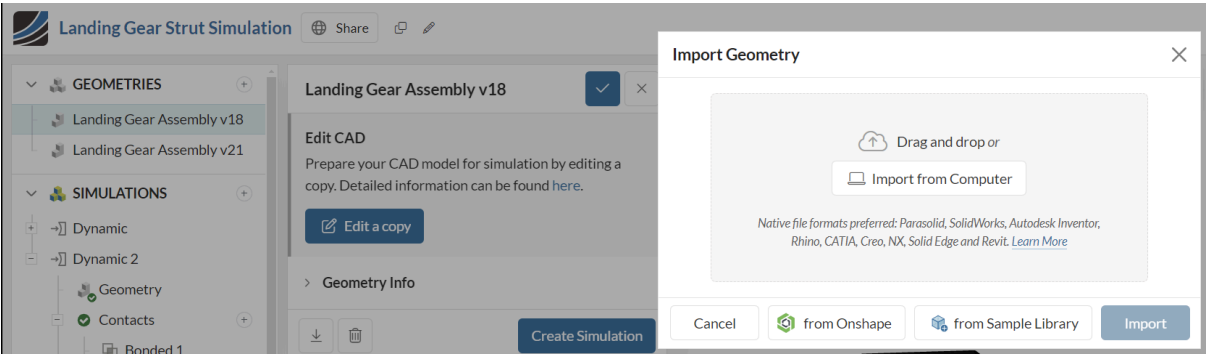
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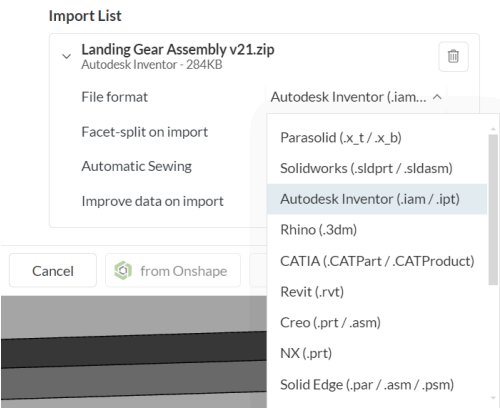
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1. Cad Import:



Under Geometries, import the CAD file as a local format type

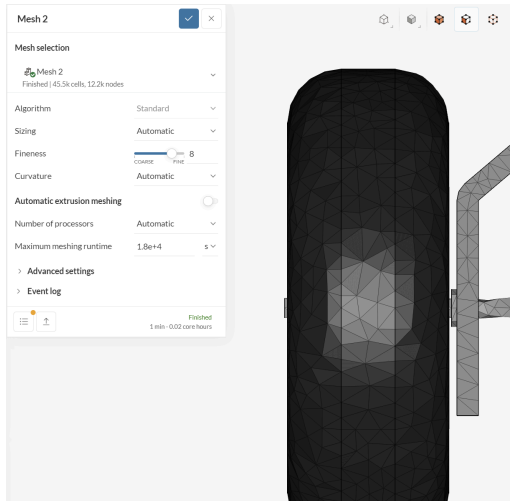


For Fusion 360, save and import as Autodesk Inventor (.iam/.ipt) file type

2. Simulation Settings and Setup

Meshing

After importing your CAD file, you will need to generate a mesh. A mesh breaks down the geometry of your model into smaller, discrete elements that the simulation solver uses to calculate physical behaviour.

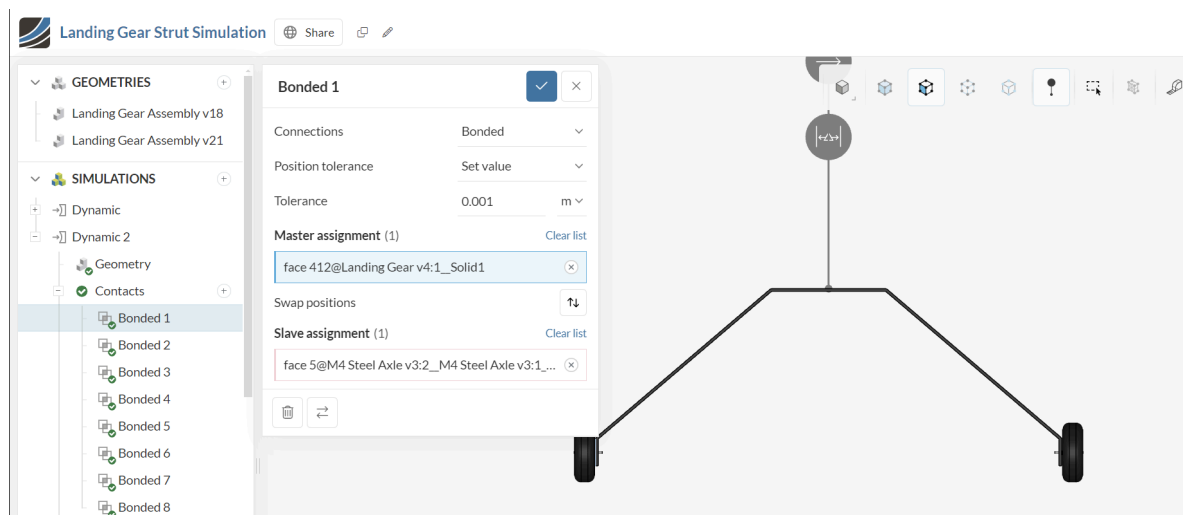


Meshing - Simscale

Contacts

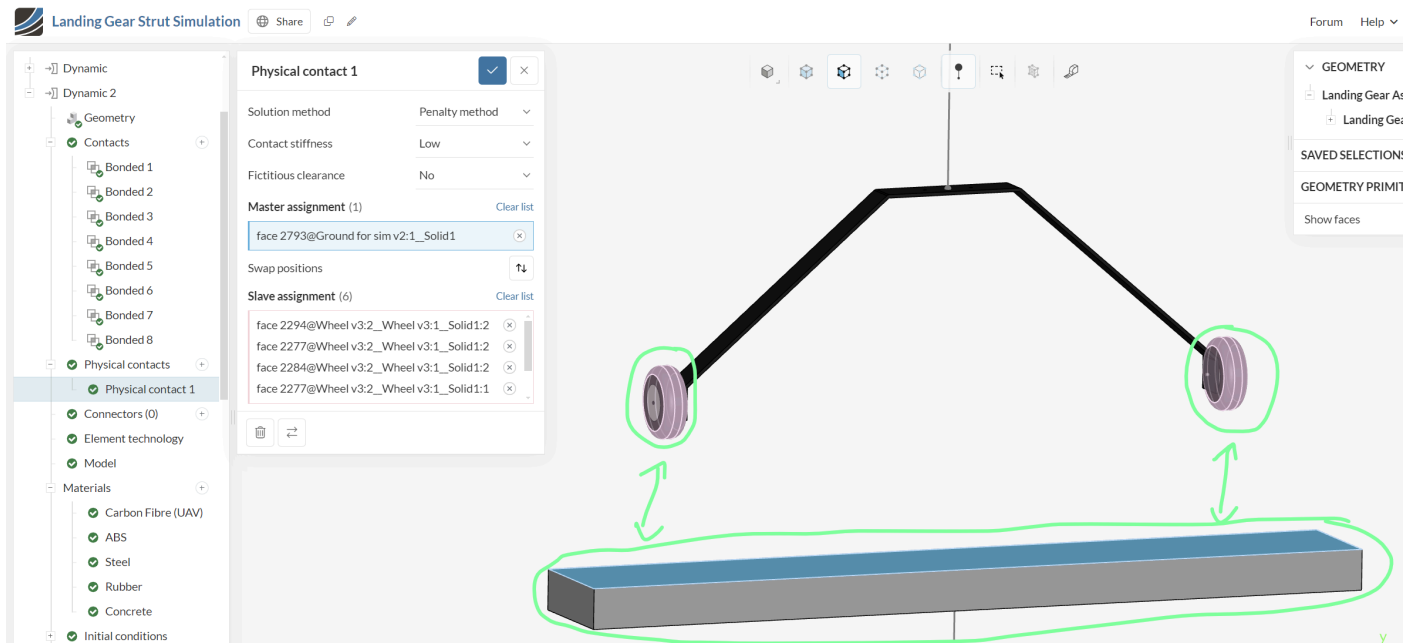
Contacts act as mates. Contacts are usually transferred automatically from CAD software joints.

There are 2 types of contacts; Bonded and Sliding.



Physical Contacts

Add physical contacts between faces that will undergo an impact with each other.



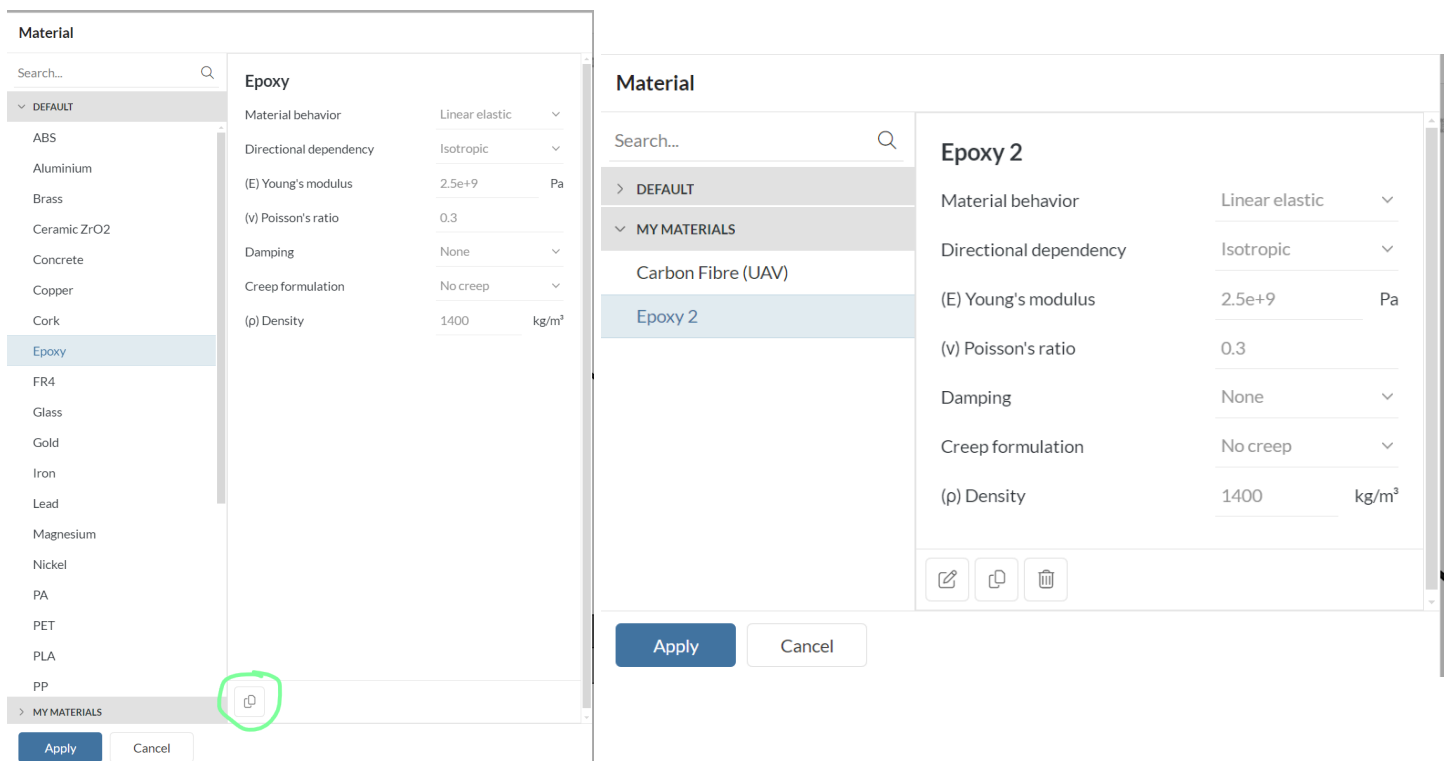
Here, the wheels and the top face of the model ground have a physical contact with the ground being the “master assignment” and the wheels are “slave assignments”.

Material

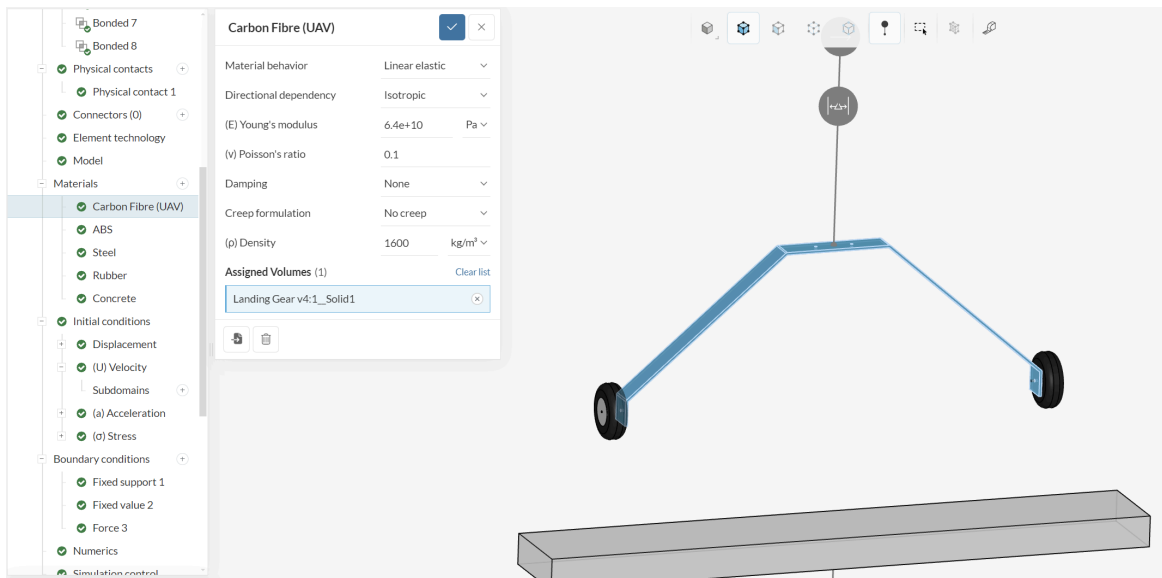
Assign materials for each part.

You can assign materials from Simscale’s pre-loaded library or create your own.

To create your own material/modify properties, you need to copy a pre-existing material from the library and it will appear under “My Materials”. From there, you can edit the material characteristics.



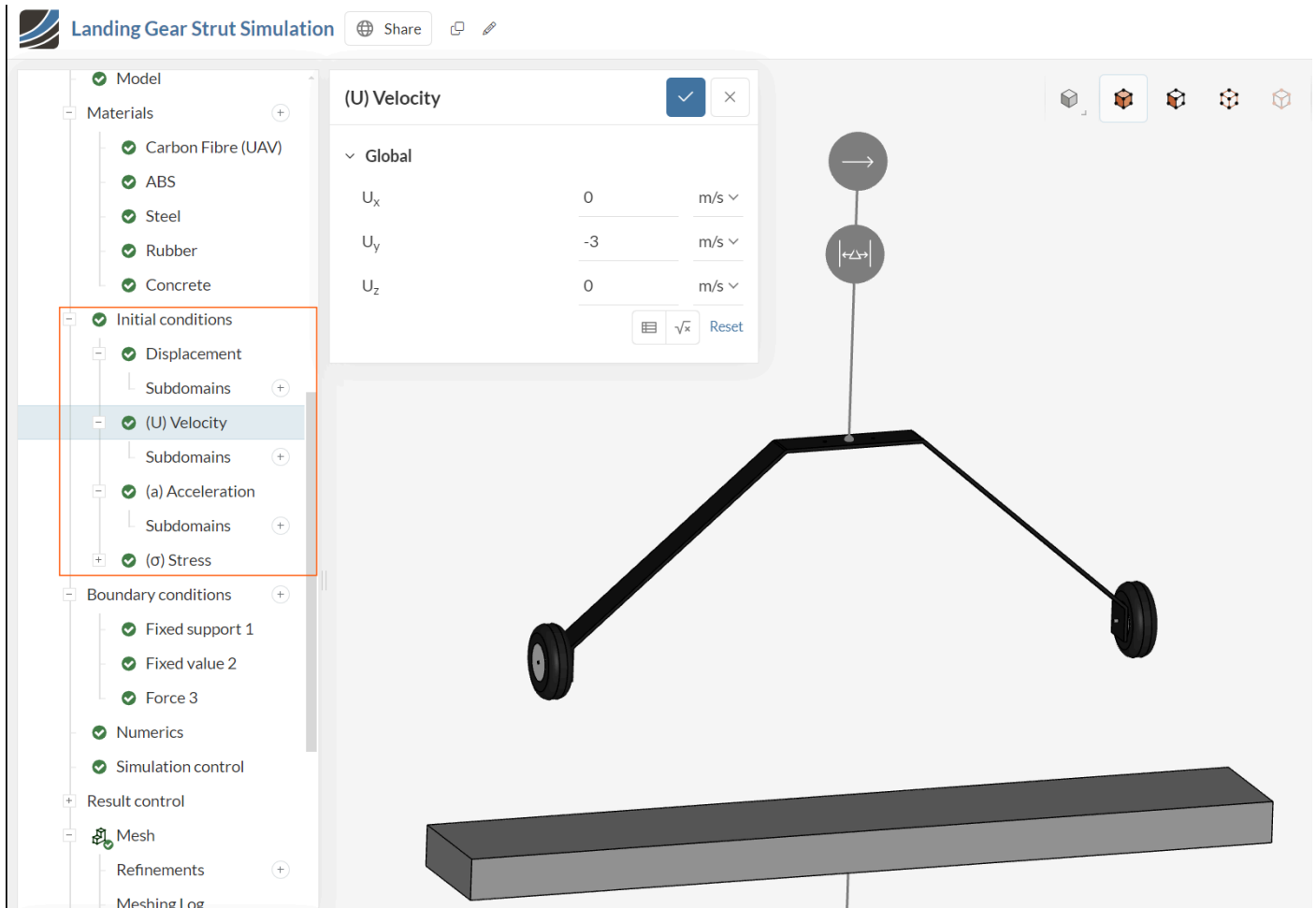
Assign the material by clicking on the object.



Materials - Simscale

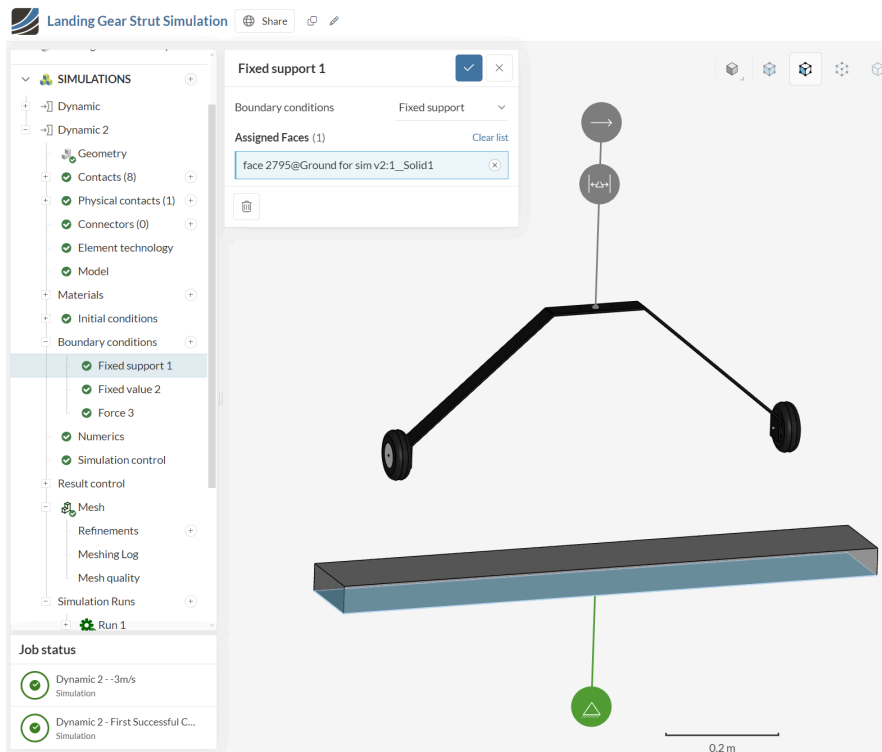
Initial Conditions

Initial conditions are the starting parameters of a simulation, defining the state of the system before any forces or loads are applied. They set the baseline for how the simulation evolves over time.



Boundary Conditions

Boundary conditions define how a simulation interacts with its environment at the edges or boundaries of the domain. They specify the constraints, forces, or flows at these boundaries.



Boundary Conditions - Simscale

Simulation Control

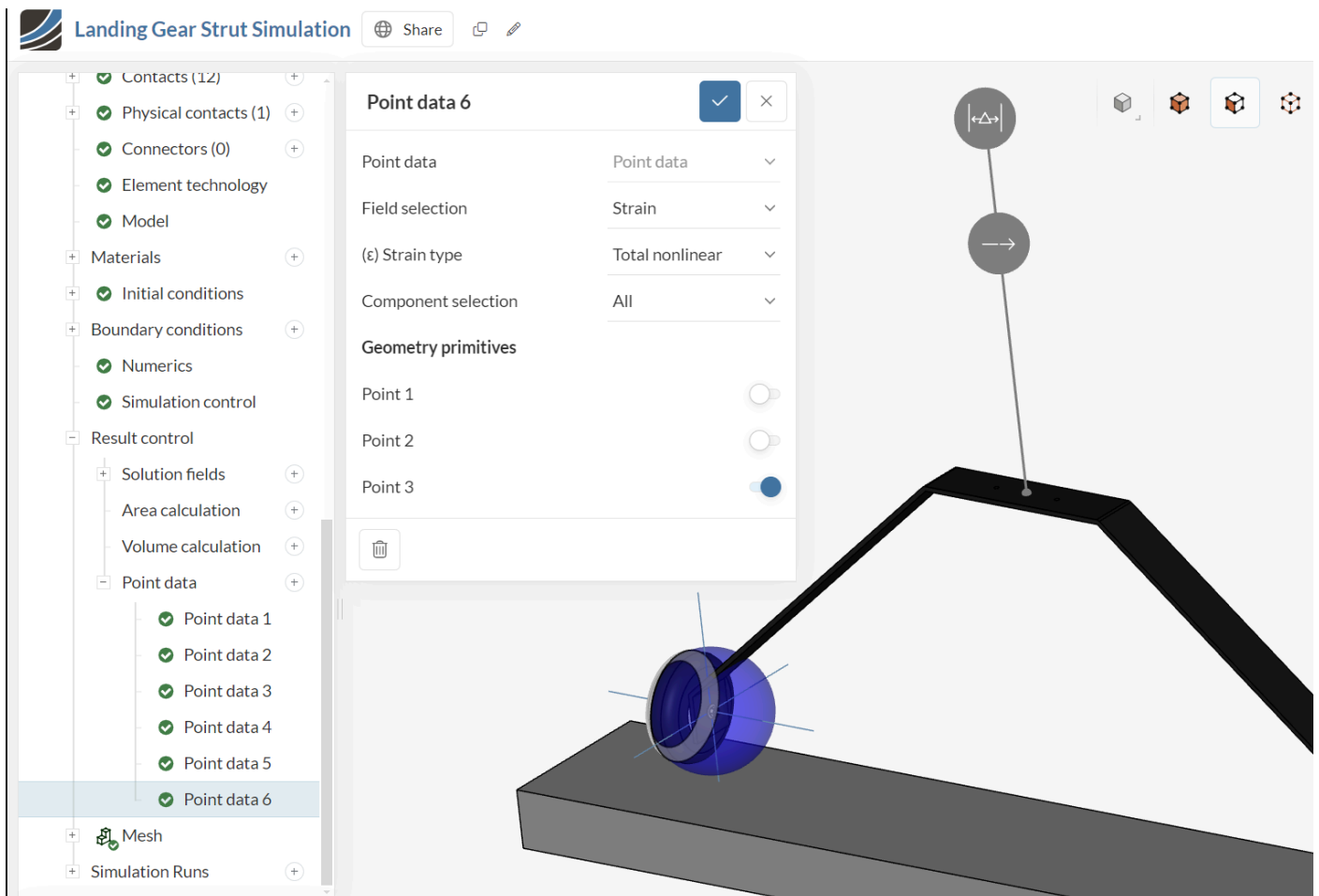
Simulation control defines how the simulation is executed, managing the time steps, accuracy, and computational resources.

1. **End Time/Simulation Duration:** Specifies how long the simulation runs in real or simulated time.
2. **Time Step Size:** Defines the increments in time for each calculation. Smaller time steps increase accuracy but require more computational resources.
3. **Maximum Number of Iterations:** Limits the total number of steps for iterative solvers, helping prevent excessive computation.
4. **Convergence Criteria:** Sets thresholds for when the solution is considered stable or accurate enough to stop the simulation.
5. **Solver Controls:** Adjusts how the numerical solver works, including methods for solving fluid flow, heat transfer, or structural behaviour.
6. **Automatic vs. Manual Settings:** Some simulations allow automatic adjustment of time step size, while others can be manually tuned for precision or speed.

Simulation Control - Simscale

Result Control

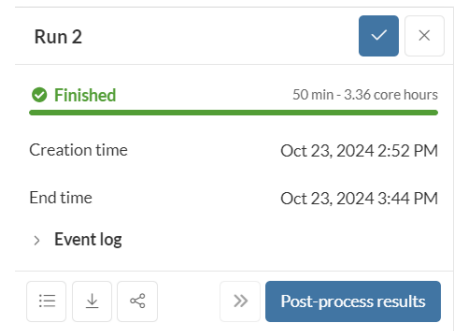
Result Control in SimScale allows you to extract specific data such as forces, displacements or stresses at selected points, surfaces, or volumes during the simulation. It helps you focus on key metrics by defining probes, surface data, or force plots, ensuring you capture relevant information. After running the simulation, you can visualise and export the data for further analysis.



[Result Control - Simscale](#)

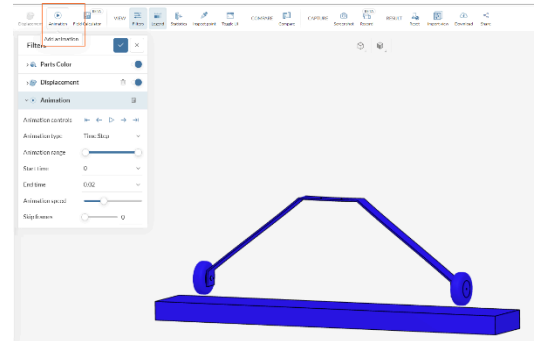
3. Post-Process Results

After the simulation is completed, you can enter the post-process results. Post-processing in SimScale refers to the analysis and visualisation of simulation results after the simulation run is completed. You can see various outputs like stress, displacement, temperature, pressure, and other physical quantities.



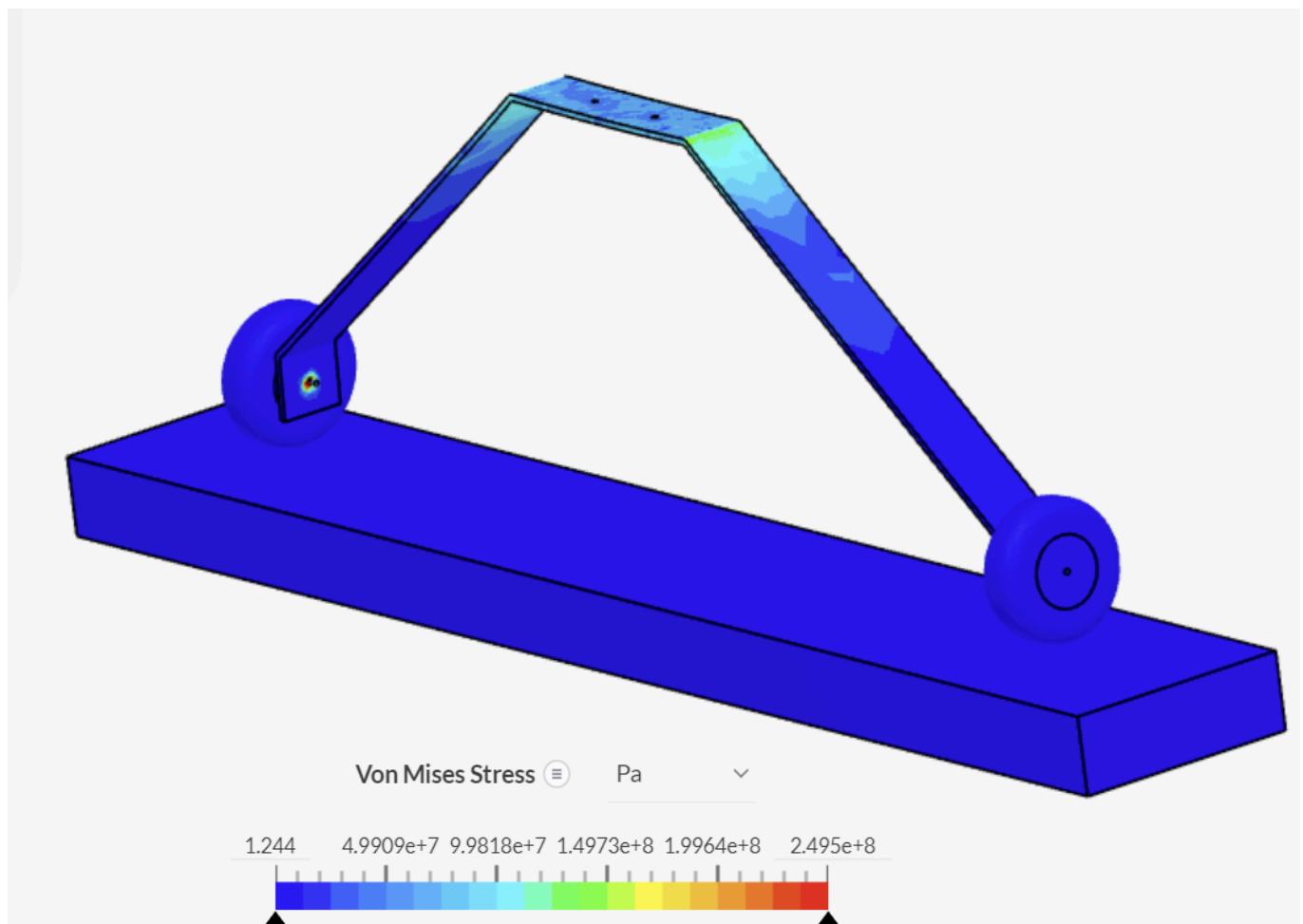
Useful Features:

1. **Animation:** Visualise the dynamic behaviour and simulation of the model.
2. **Probes:** Extract specific data, such as stress or displacement at particular points
3. **Force/Stress/Displacement Plots:** Create graphs from tracked metrics over time such as forces, stresses and displacement.



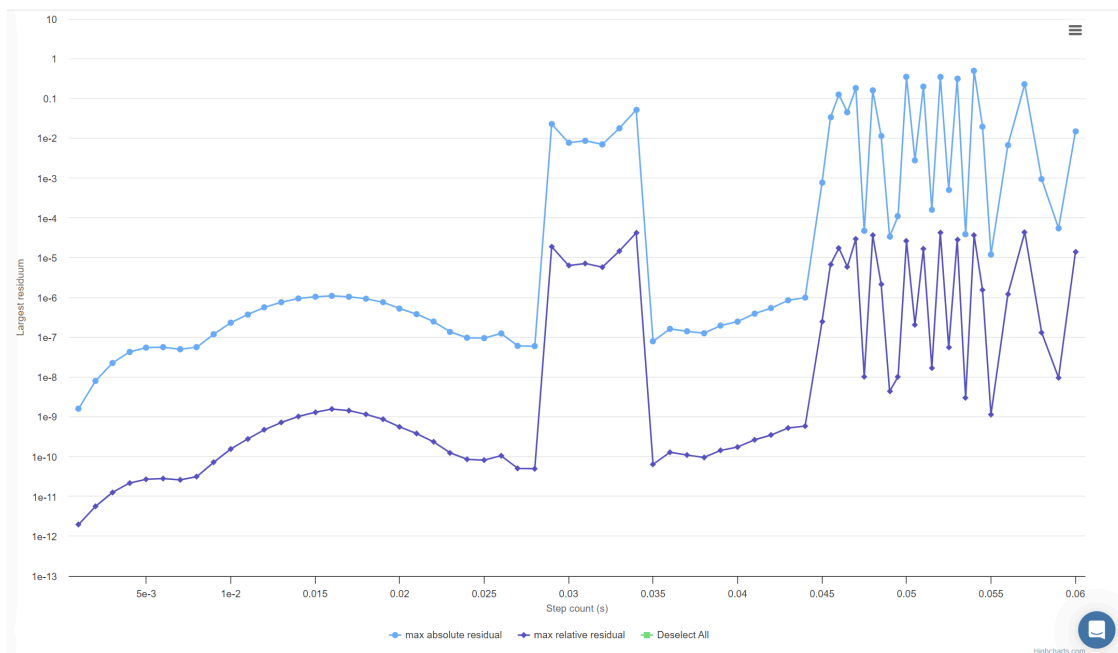
Von Mises Stress:

If the von Mises stress of a material under load is equal to or greater than the yield limit of the same material under simple tension then the material will yield.



Convergence:

This is plotted on a logarithmic scale, showing the size of the residuals in the simulation. Residuals are a measure of the difference between the predicted and actual values in each step of the simulation. Smaller residuals indicate better convergence, while larger residuals suggest numerical instability or issues with the solution's accuracy.



If your step sizes are too large, the simulation run will fail due to a failure of convergence.